Digital State-Feedback Regulation of a Cart-Pendulum System

1 Introduction

The purpose of this lab is to design digital state feedback regulators for an inverted pendulum-on-a-cart system. Different designs will be compared by simulation, and the stability margins will also be compared. The best regulator will be used to regulate a hardware system.

2 Regulator Design - Week One

A linearized model for this system is given in equation (3.167) of the book, and the parameter values are shown in Table 3.9. Use Table 3.9 but replace the values of C and D as follows. Assume that the transfer function of the motor system found in Lab 1 (with motor velocity as the output) is $\beta/(s+\alpha)$. **Replace the value of D with β , and replace C with α.** For the hardware station closest to the window, use the value $A = 23.1$ in equation (3.167). **(Note, call this variable something else in Matlab, because A is reserved for the state-space matrix of the continuous time plant model.)** For the other hardware station, use $A = 41.5$. (The second hardware station uses a shorter pendulum than the first station, and the value of A is the square of the radian natural frequency of the pendulum.) The four state variables of the plant are:

$$\begin{aligned} x_1 &= \text{pendulum position (rad)} \\ x_2 &= \text{pendulum velocity (rad/sec)} \\ x_3 &= \text{motor position (rad)} \\ x_4 &= \text{motor velocity (rad/sec)} \end{aligned}$$

Use a sampling interval of $T = 0.01$ seconds. We would like to compare two different regulators, each designed to achieve a different set of closed-loop poles. For each set of closed-loop poles, find the associated stability margins.

1. The first set of closed-loop poles are the roots of a 4th-order Bessel polynomial, scaled by T_S. In this design, T_S is not directly specified. Rather, you have to find, through simulation, the **smallest** value of T_S that achieves the following constraint. An initial condition $\mathbf{x}(0)$ (shown below) **results in a control signal satisfying** $-5 \le u(t) \le 5$. This initial condition vector represents the pendulum leaning at an initial angle of 0.17 radians (about 10 degrees).

$$\mathbf{x}(0) = \begin{bmatrix} 0.17 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

2. The second set of desired closed-loop poles uses sufficiently damped plant poles, with Bessel poles scaled by T_S used to get to $n = 4$ desired poles that satisfy the constraint given above.

3 Simulink Simulation - Week One

The regulators are implemented as shown in Fig. 1. Note that this diagram uses matrix gain blocks implementing the following vectors: $[1 \quad 0 \quad 0 \quad 0]$, $[0 \quad 0 \quad 1 \quad 0]$. Passing $\mathbf{x}(t)$ through these gain vectors produces the signals $x_1(t)$ and $x_3(t)$, respectively, which can be viewed on

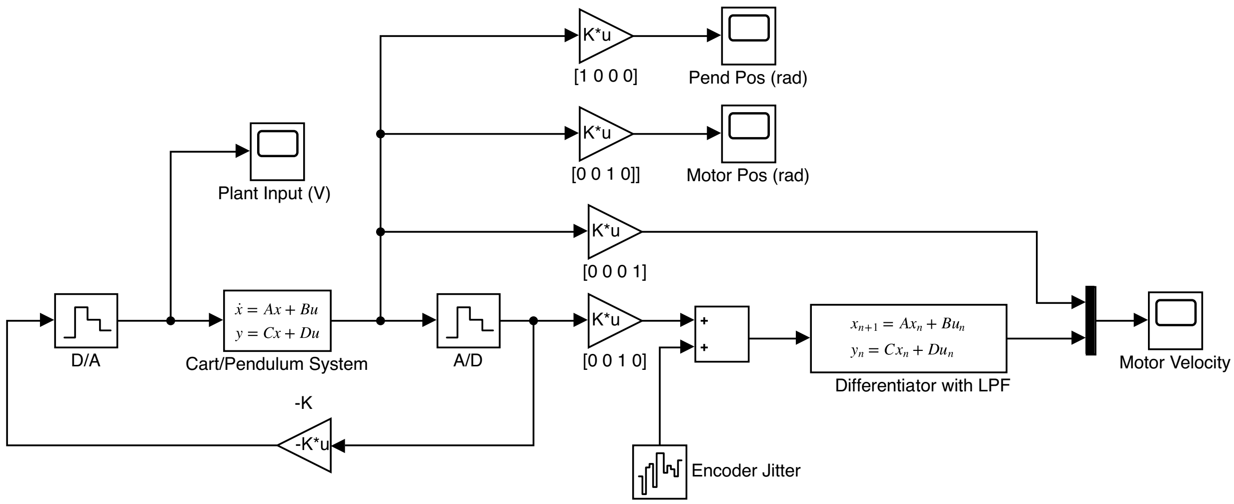

Figure 1: State-feedback regulator for the cart/pendulum system

Simulink scopes. **Use Simulink ZOH blocks (found under *Discrete*) to model both the D/A and A/D converters (set sample time to T). The Encoder Jitter block is actually a Band-Limited White Noise block found in *Sources*. Set the noise power to be 0.002 and the sampling time to T. The block that combines the two signals going into the Motor Velocity scope is a Mux, which is found in *Signal Routing*.** [Note the default colors when a scope shows more than one signal: yellow for the first (top) signal and blue for the second (bottom) signal.] Set the sampling interval to T seconds. The model in Fig. 1 simulates all of the state variables with a continuous-time state-space model and then samples them with A/D converters to perform digital state feedback.

In the hardware system we have direct measurement of sampled position signals $x_1[k]$ and $x_3[k]$. Neither of the velocity signals ($x_2[k]$ or $x_4[k]$) are measured; they must be obtained from the measured signals. In order to obtain these velocity signals we will use a digital differentiator combined with a digital lowpass filter. The filter is called a **"differentiator with lowpass filter"** and is implemented with a **discrete-time state-space model** $(\mathbf{A}_f, \mathbf{B}_f, \mathbf{C}_f, \mathbf{D}_f)$. The simulation diagram above sends the simulated analog motor velocity $x_4(t)$ to a scope, as well as the digitally differentiated motor position signal. These are compared in order to design the lowpass filter by choosing the filter speed f. A small value of f (slow filter) filters noise well but does not provide a good estimate of velocity. A large value of f provides a good estimate of motor velocity but does not filter noise well. Choose the value of f that gives a good trade-off.

```
Tsf = Ts/f; % filter settling time; choose f to be greater than or equal to 5
bwfp = (s1+j*s1)/Tsf; % 2nd-order Butterworth filter pole scaled to Tsf-sec settling time
den = poly([bwfp conj(bwfp)]);
num = [den(end) 0]; % numerator is the constant term in the denominator polynomial
vfilter_a = tf(num,den); % analog differentiator with lowpass filter
vfilter_d = c2d(vfilter_a,T,'tustin'); % digital differentiator with lowpass filter
[Af,Bf,Cf,Df]=ssdata(vfilter_d); % convert vfilter_d to state-space model
```

4 Hardware - Week Two

Create a Simulink model to control the hardware system. You will need to measure the four state variables in order to implement a state-feedback regulator. None of the state variables will be measured with an A/D converter. Instead, pendulum position and motor position will be measured with optical encoders. The corresponding velocities will be obtained by digitally differentiating and filtering the position signals.

5 Write Up

The report should contain results from both the simulation and hardware parts of the lab. Use the format described in the handout, **ELE459 - Lab Reports**, which is available on the course web site under Handouts. In particular, include the following information in your report:

1. Describe the formulation of the digital state-feedback regulator design to achieve the desired specifications while satisfying the given constraints.

2. Comment on the choice of sampling interval, the choice of closed-loop poles, the choice of settling time, and the stability margins of each design.

3. Perform a simulation of the cart/pendulum system that can be compared with the measured data. The appropriate initial condition vector to simulate a tap on a balanced pendulum (as was done in the lab) has a nonzero initial velocity, as follows: $x_2 = 0.17$ with all other state variables zero. That is, `x0=[0;0.17;0;0]`. **Note that this initial condition is not the same one that was used to choose the settling time.** Compare this simulation result with a plot from the hardware system.

4. Give details about the hardware set up. In particular, describe the issues regarding the encoder measurement of pendulum position and describe how the hardware system was initialized.

5. Explain how the control system was implemented on the lab computer, what the difficulties were, and how the problems were solved. In particular, describe the problems with noisy state-variable measurements and how this affected the final design.